MIL-T-81311A(AS)
15 February 1980
SUPERSEDING
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7 September 1965

MILITARY SPECIFICATION

TRAILER, AIR LAUNCHED WEAPONS, ROUGH TERRAIN, A/M32K-4A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of one type of flat bed, pneumatic tired, rough terrain trailer for air launched weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof - for Packaging and Sealing.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1730

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MILITARY (continued)

MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification for.
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-810	Environmental Test Methods.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

Data List 67A219J1 Rough Terrain A/M 32K-4A Trailer .Assembly (And all documents listed thereon).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) the contractor shall furnish one sample unit for first article inspection and approval (see 4.2 and 6.2).

3.2 Construction. The trailer shall be manufactured in strict accordance with Naval Air Systems Command Data List DL67A219J1 and all documents listed thereon.

3.2 Performance Characteristics. The trailer covered by this specification shall meet the following performance characteristics.

3.3.1 Bed - The bed of the trailer shall be furnished with a non-skid surface in accordance with applicable drawings.

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When the cradle tie-down fittings described in 3.3.2 are removed, the bed of the trailer shall be free of any obstructions or protuberances.

3.3.2 Cradle tie-down fittings. - As shown on applicable drawings, 12 cradle tie-down fittings shall be supplied with each trailer. These mechanical devices shall slide in the tracks provided in the trailer bed and are designed to hold certain weapons cradles rigidly to the bed.

3.3.3 Brakes.

3.3.3.1 Service. - As shown on applicable drawings, air over hydraulic brakes shall be placed on the two rear wheels of the trailer. Power is supplied by the prime mover.

3.3.2 Parking. - Mechanical parking brakes, as shown on appli-cable drawings shall be utilized on the trailer.

3.3.4 Suspension. - As shown on applicable drawings, suspension is of the independently sprung type with one multiple leaf spring at each wheel. Each axle shall be provided with bump stops to insure that maximum deflection to stop is three inches.

3.3.5 Cramping stops. - Stops shall be furnished on the trailer chassis to restrict the cramping angle to a maximum of 40 degrees each side of center.

3.4 Mobility. - Trailers supplied under this specification shall be capable of meeting, without damage, the requirements listed below. In the event any conflict arises between the aforementioned specification and this document, the requirements herein stated shall apply.

Requirement	Test Paragraph
(a) Towing loaded trailer: .	
(1) on improved concrete between 35 and 45 mph for 50 miles.	4.4.2
(2) on improved gravel at 25 mph for 25 miles.	4.4.3
(3) across open fields and broken terrain at 5 mph for 25 miles.	4.4.4
(4) through figure eights at eight mph and demonstrating turning radius of not greater than 15 ft.	**4.4.5**

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Requirement	Test Paragraph
(5) across 20 percent grade five times in each direction.	**4.4.6**
(b) Braking loaded Trailer:	
(1) on 20 percent grade from - 20 mph. within 30 ft.	4.4.7.1
(2) on level concrete from 20 mph within 30 ft.	4,4.7.2
(3) skidding wheels when towed with parking brakes applied.	4.3.6
(4) parking on 20 percent grade with trailer facing uphill	4.4.7.3.1
(S) parking on 20 percent grade with trailer facing downhill	4.4.7.3.2
(6) brakes lock when towing force removed.	4.3.6
(c) Drawbar pull and static resistance:	
(1) Trailer shall roll before 225 pounds is applied through drawbar.	4.3.5

3.5 Environmental requirements. - Trailers supplied under this specification shall be capable of meeting, without damage, the following environmental criteria:

Requirement	Test Paragraph
(a) High temperature - 160°F for eight hours.	4.5.1
(b) Low temperature - minus 65°F for eight hours.	4.5.2
(c) Humidity - Method 507.1, Procedure 1 of MIL-STD-810.	4.5.3
(d) Sand and dust - Method 510.1, Procedure 1 of MIL-STD-810.	4.5.4
(e) Fording - 30 minutes immersion in water.	4.5.5

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3.5 Workmanship. - The workmanship displayed in fabrication and assembly of the trailer shall be such as to assure, within design limitations, ability of the' trailer to meet its performance requirements under all applicable environmental conditions. Unauthorized repair, welding, loose rivets, heavy burrs, indiscriminate placement of fasteners, and parts assembled by introduction of high stresses not prescribed by design, are typical signs of inferior workmanship. The standards of workmanship exhibited in any approved first article sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed - requirements.

4.2 Classification of inspections. - The examination and testing of trailers supplied under this specification shall be as follows:

(a) First article inspections
(b) Quality conformance inspections

4.2.1 First article inspection - The first article sample and its related components shall be subjected to all examinations and tests of this specification. Unless otherwise specified, these examinations and tests shall be performed by the contractor.

4.2.2 Quality conformance inspection. - Examination and testing of each trailer for acceptance shall consist of those procedures specified in 4.3.

4.3 Quality conformance examination and testing.

4.3.1 Inspection conditions. - Unless otherwise specified, trailers shall be tested under the following conditions:

(a) Temperature - 77 ± 18 degrees Fahrenheit

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(b) Humidity - 90 percent, maximum

(c) Barometric Pressure - Local Standard

4.3.2 Examination. - The trailer shall be inspected to determine compliance with the requirements specified herein with respect to drawings, dimensions, materials, clearances, markings, and workmanship.

4.3.3 Cradle tie-down fittings examination. - It shall be determined that 12 cradle tie-down fittings have been furnished with each trailer and that these fittings slide freely in the channel provided. The Government will, at its option (see 6.2), furnish the contractor with certain weapon cradles to determine that the fittings will, in fact, secure the cradles and that the fittings are adjustable across the range indicated on applicable drawings.

4.3.4 Suspension and steering examination. - In addition to the examination of 4.3.2, it shall be determined that the bump stops (see 3.3.4) and the cramping stops (see 3.3.5) are present and in a location which will not preclude trailers from meeting requirements of the aforementioned paragraphs.

4.3.5 Drawbar pull and static resistance test. - A scale shall be placed between the prime mover and tbe trailer which has been loaded with 8,000 pounds. The prime mover shall apply a gradual towing force through the scale to the loaded trailer with its brakes released and with the trailer on level concrete. The trailer shall roll before the scale registers 225 pounds. This process shall be repeated three times.

4.3.6 Skidding. - The trailer, loaded with 3,000 pounds (see 4.4.1) shall be stopped on level concrete and the parking brakes applied. The wheel shall skid, not roll, when the prime mover tows the trailer five feet. It shall also be determined that the service brakes lock when the pneumatic line from the prime mover is accidentally disconnected.

4.4 Mobility testing. - Unless otherwise specified in the contract or order, sample trailers supplied as specified in 4.2.1 shall be prepared and tested as described in the following paragraphs:

4.4.1 Preparation. - The trailer shall be loaded with a weight of 8,000 pounds with the combined center of gravity located equidistant between the axles on the longitudinal centerline and eight inches above the trailer bed. The loaded trailer shall then be connected to a military type vehicle. The electrical cable and pneumatic hoses of the trailer shall be attached to the prime mover.

NOTE: NO adjustment or servicing of any type shall be performed on the trailer during any of the following road tests.

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4.4.2 Towing on improved concrete surfaces. - The loaded trailer shall be towed along improved concrete surface roads at an average speed between 35 and 45 miles per hour for a distance of 50 miles. (Do not tow faster than 45 mph at any time.) The towing characteristics of the trailer shall be noted.

4.4.3 Towing on improved gravel surfaces. - The loaded trailer shall be towed along improved gravel surface foads at an avarage speed of 25 miles per hour for a distance of 25 miles. Twenty-five sudden stops from 25 miles per hour shall be made during the test and thw towing and braking characteristics noted.

4.4.4 Towing across open fields and broken terrain. - The loaded trailer shall be towed across open fields and generally broken terrain with spaced bump obstacles at an average speed of five miles per hour (do not tow faster than ten mph at any time) for a distance of 25 miles. Twenty-five sudden stops from five miles per hour shall be made during the test and the towing and braking characteristics noted.

4.4.5 Towing through figure eights. - The loaded trailer shall be towed through 25 figure eights at eight ± one mile per hour. The outside wheels of the trailer shall circumscribe a circle 38 feet or less in diameter. The towing and turning characteristics and any signs of the trailer reaching an unstable or tipping condition shall be noted.

4.4.6 Towing across 20 percent grade. - The loaded trailer shall be towed across a 20 percent (11.3 degrees) grade five times in each direction and the towing characteristics noted. The trailer shall show no signs of tipping during this test. No damage to the trailer or compenents shall become evident either during or after this test.

4.4.7 Brake tests.

4.4.7.1 Braking on 20 percent grade, - The loaded trailer shall be towed down a minimum grade of 20 percent (11.3 degrees) and brought to a stop with the service brakes from 10 miles per hour. The braking characteristics shall be noted.

4.4.7.2 Braking from 20 miles per hour. - The loaded trailer while being towed at 20 miles per hour on level concrete (after slowing from 25 miles per hour to eliminate driver reaction time) shall be brought to a stop within 30 feet. The braking characteristics shall be noted.

4.4.7.3 Parking on 20 percent grade.

4.4.7.3,1 The loaded trailer shall be towed to a position where the front of the trailer is facing up a 20 percent (11.3 degrees) grade and the parking brakes applied. The towing forces shall be removed and the trailer shall hold its position for a minimum time of 30 minutes.

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4.4.7.3.2 The loaded trailer shall be towed to a position where the front of the trailer is facing down a 20 percent (11.3 degrees) grade and the parking brakes applied. The towing forces shall be removed and the trailer shall hold its position for a minimum time of 30 minutes.

4.4,8 Examination. - Upon completion of the foregoing tests, the trailer shall be examined. The trailer shall not have suffered any damage which is considered sufficient to preclude its meeting any performance or operational requirements of this specification.

4.5 Environmental testing. Unless otherwise specified in the contract or order, sample trailers supplied as specified in 4.2.1 shall be subjected to the following environmental testing.

4.5.1 High temperature. - The trailer shall be loaded with 8,000 pounds as described in 4.4.1 and exposed to an ambient temperature of 160 degrees Fahrenheit for eight hours. At the end of the eight hour exposure and while still at 160 degrees, the trailer shall be rolled to assure function of moving parts and brakes. The temperature of the room shall be returned to 80 ±10 degrees Fahrenheit and the trailer examined.

4.5.2 Low temperature. - The loaded trailer shall be exposed to an ambient temperature of minus 65 degrees Fahrenheit for eight hours. At the end of the eight hours exposure and while still minus 65 degrees, the trailer shall be rolled to assure function of moving parts and brakes. The temperature of the room shall be returned to 80 ±10 degrees Fahrenheit and the trailer examined.

4,5.3 Humidity. The trailer shall be unloaded and subjected to the Humidity Test of Method 507.1, Procedure I, M1L-STD-81O. At the conclusion of the ten cycles, the trailer shall be returned to a normal humidity environment and examined. The trailer shall be towed a short distance to assure function of moving parts and brakes.

4.5.4 Sand and dust. - The loaded trailer shall be subjected to the Sand and Dust Test of Method 510.1, Procedure I, MIL-STD-81O. The air velocity shall be between 100 and 500 feet per minute. At the conclusion of the test, the trailer shall be returned to a normal environment and examined. The trailer shall be towed a short distance to assure function of moving parts and brakes.

4.5.5 Fording. - The trailer, loaded as described in 4.4.1 shall be towed into a 30 inch deep fording tank. After 30 minutes immersion, the trailer shall be removed from the tank and attached to a prime mover, The trailer shall then be towed at 15 miles per hour and the brakes applied. The braking characteristics shall be noted.

5. PACKAGING

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5.1 Preservation packaging. - Preservation packaging shall be either Level A or Level C, as specified (see 6.2).

5.1.1 Level A (mobile).

5.1.1.1 Cleaning and drying. - The trailer shall be cleaned and dried in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with Procedure C-1. Drying shall be accomplished in accordance with Procedure D-1 or D-4, or a combination of both.

5.1.1.2 Preservation application,

5.1.1.2.1 Unpainted bearing surfaces. All unpainted bearing surfaces shall be coated with grease conforming to MIL-G-23827, and wrapped with grease-proof barrier material conforming to MIL-B-121, Type II, Grade A, Class 2, secured in place with tape conforming to PPP-T-60.

5.1.1.2,2 Unpainted non-bearing surfaces. All unpainted non-bearing surfaces shall be treated with preservative compound conforming to MIL-C-16173, Grade 1.

5.1.1.2.3 Tires, tubes, wheels, hubs, and axles. - Tires and tubes shall be mounted on the wheels and shall be inflated to five pounds above recommended service pressures. Interior bearing surfaces of the hub and the bearing surfaces of the hub and the bearing surfaces of the axle shall be coated with grease conforming to MIL-G-23827.

5.1.1.2.4 Pneumatic system and cable ends. - Openings into the pneumatic system and the ends of electrical cables shall be covered with grease-proof barrier material conforming to MIL-B-121, Type II, Grade A, Class 2, secured with tape conforming to PPP-T-60.

5.1.1.2.5 Technical publications. - Any technical publications supplied with the trailer shall be packaged in accordance with Method lC-1 of MIL-P-116.

5.1.2 Level C (mobile). - The trailer shall be prepared for shipment in accordance with the supplier's standard commercial practice, in a manner acceptable to common carriers for safe transportation to point of destination at the lowest transportation rate for the commodity. This level is intended to provide adequate protection for immediate use domestic shipments, from supply source to the first receiving activity.

5.2 Marking. In addition to any special markings required by the contract or order, the basic trailer and any interior packages shall be marked in accordance with MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The vehicle described herein is intended for use by the Marine Corps in transporting air launched weapons, primarily at expeditionary airfields, in an extremely rough terrain environment.

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6.2 Ordering data. - Procurement documents should include the following:

(a) Title, number, and date of this specification.

(b) Whether or not a first article sample is required.

(c) Number of Trailers which will constitute the first article sample.

(d) Information pertinent to the site of first article testing (see 4.2.1).

(e) Whether or not test reports are required (see 4.2.1).

(f) Whether or not weapon cradles will be supplied for acceptance testing (see 4.3.3).

(g) Level of preservation and packaging required (see Section 5).

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